

A USER INTERFACE OF SUPER-GAUSSIAN

Super-Gaussian provides three main user interfaces. Two of them are designed for direct GUI-based scene control and editing, while the third supports NLI, allowing users to follow the conversation and inspect intermediate results, such as intent parsing and open-vocabulary queries. Specifically, the scene control panel in Figure 1 (a) allows users to edit visualization attributes of editable units, including color, opacity, and lighting parameters. The 3D prompt panel in Figure 1 (b) supports interactive region selection and labeling. Users can adjust brush width, choose selection methods (RW, cluster, or point), and modify the color and opacity used to highlight selected regions. It also provides GUI-based text labeling and options to either save the selection as a new editable unit in Figure 1 (a) or reset the current selection. The NLI interface in Figure 1 (c) provides a conversational interface where users can review voice commands and system responses in text form. It also includes an Agent Commands panel for inspecting intermediate agent operations, such as intent parsing and execution steps, and a Query Result panel for displaying the results of open-vocabulary queries.



Fig. 1: User interface of Super-Gaussian in the VR environment. (a) Scene control panel for GUI-based Scene editing. (b) 3D prompt panel for brush interaction, selection control, and semantic labeling. (c) NLI interface for voice-command-based volume exploration and scene editing.

B EDITABLE GAUSSIAN FEATURES

Editable Gaussian primitives are parameterized by both geometric and appearance-related attributes. Geometric attributes include anisotropy, scale, normal directions, and principal axes, which describe the local structure and orientation of each Gaussian. Appearance-related attributes include opacity and offset color, which control the visual representation of the volume. As illustrated in Figure 2, these attributes exhibit distinct spatial distributions across the volume. In particular, geometric attributes tend to form coherent patterns aligned with underlying structures, while appearance-related attributes often show less structural consistency. These properties are important for our method, as structure-aware attributes (e.g., scale, normal, and principal axes) are used to guide Super-Gaussian clustering and enable more reliable region selection.

C RENDERING EFFICIENCY COMPARISON

Table 1 summarizes the rendering efficiency of our editable Gaussian representation and compares it with a naive Unity-based DVR baseline under the same VR benchmarking protocol. FPS is measured by traversing viewpoints along a scripted 360° camera trajectory after a warm-up rotation, and computed from per-frame timing using the total measured frames and rendering time. For the DVR baseline, we used

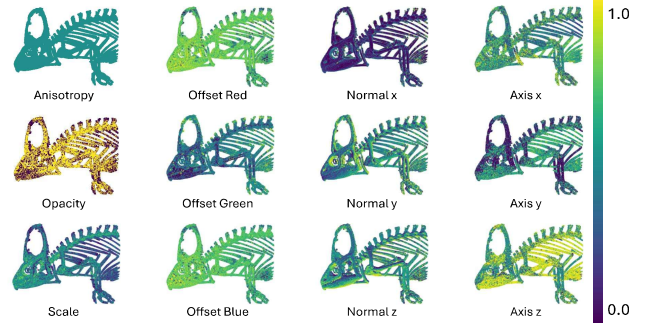


Fig. 2: Gaussian attributes of the chameleon dataset color-mapped onto the 3D scene to illustrate its spatial distribution, including anisotropy, opacity, scale, offset color, normal directions, and principal axes.

the open-source UnityVolumeRendering² package in its direct volume rendering mode with Phong lighting. The sampling rate multiplier was set to 1.0 (the default), and early ray termination was enabled, which terminates ray marching once the accumulated opacity exceeds 0.996. As shown in the table, DVR performance degrades substantially as the volume resolution increases. While DVR remains feasible for smaller datasets, its frame rate drops noticeably even for 512³ volumes, and datasets larger than beetle could not be rendered due to Unity’s 3D texture allocation limit. In particular, for the 512³ datasets (aneurysm, bonsai), the minimum DVR frame rate falls below 5 FPS in some views, indicating severe view-dependent performance fluctuation. Even when the average FPS exceeds 60, such extreme drops would likely cause substantial discomfort in VR and significantly degrade the interactive user experience. These results highlight that naive Unity-based DVR scales poorly with increasing volume resolution and, in practice, requires additional techniques such as parallel rendering, out-of-core processing, foveated rendering, or downsampling to support large-scale volumetric data.

Table 1: Rendering efficiency of our editable Gaussian representation vs. DVR. OOM stands for out of memory.

dataset	volume resolution	volume size	ours avg (FPS)	DVR avg (FPS)	DVR min (FPS)	DVR max (FPS)
lobster	602×648×157	234 MB	119.92	119.88	100.28	139.44
aneurysm	512×512×512	512 MB	120.01	75.18	3.95	131.70
bonsai	512×512×512	512 MB	119.88	66.92	0.84	170.66
beetle	864×864×494	1.37 GB	120.01		OOM	
vortex	1,000×1,000×1,000	3.73 GB	118.10		OOM	
Nyx	1,024×1,024×1,024	4 GB	118.00		OOM	
chameleon	1,024×1,024×1,080	4.22 GB	119.93		OOM	

D GEODESIC SLIC ABLATION

We conduct an ablation study on Super-Gaussian construction to evaluate the effect of distance formulation in our SLIC-based clustering. All experiments are performed under identical settings, including the same number of clusters and shared hyperparameters such as voxel-based seed initialization, compactness, and graph construction, applied to the same optimized Gaussian sets for each dataset. As Super-Gaussians are designed as an intersection-oriented over-segmentation rather than a final semantic segmentation, we evaluate clustering quality qualitatively. We fix the feature representation to our proposed geometric feature set and vary the distance formulation. We compare geodesic distance with feature similarity against Euclidean distance with and without feature contribution, where the latter corresponds to a purely spatial SLIC-style baseline. As shown in Figure 3, Euclidean distance-based clustering tends to group spatially adjacent Gaussians regardless of structural continuity. For example, in the chameleon dataset, clusters fail to follow the rib structures and instead merge neighboring regions across ribs. In contrast, our method preserves the directional continuity of the ribs. A

²M. Lavik. *UnityVolumeRendering*. GitHub repository, 2024. Accessed April 6, 2026. <https://github.com/mlavik1/UnityVolumeRendering>.

similar pattern is observed in the lobster dataset, where Euclidean distance merges adjacent regions near the curved shell tip, while our method successfully separates clusters along the shell structure.

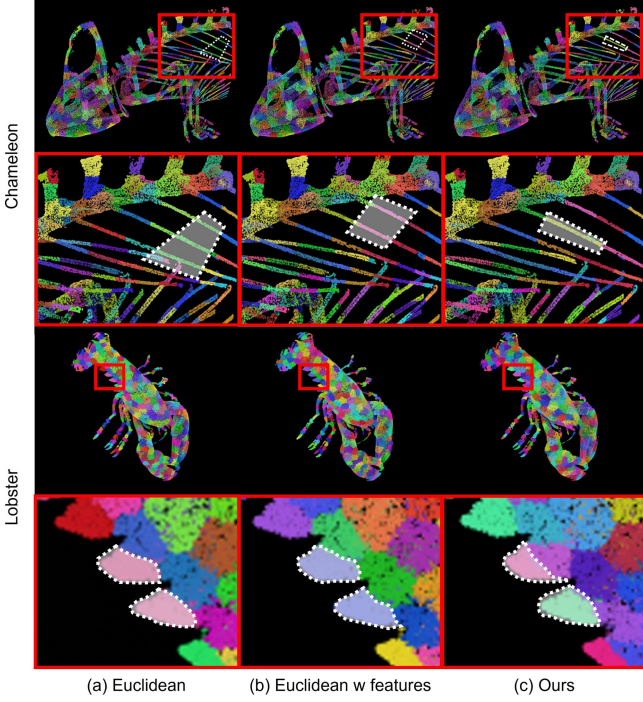


Fig. 3: Ablation on distance formulation in SLIC-based Super-Gaussian construction. The top row shows the chameleon dataset, and the bottom row shows the lobster dataset. Clusters are visualized with random colors and rendered with Gaussian centroids. (a) Euclidean distance, (b) Euclidean distance with feature similarity, and (c) ours (geodesic distance with feature similarity).

E RANDOM WALK ABLATION

We conduct an ablation study on our RW selection mode using the chameleon, lobster, and vortex datasets. Note that RW is sensitive to the choice of foreground and background seeds. Therefore, the results should be interpreted as a reference rather than absolute performance. Table 2 compares our method with Gaussian RW (i.e., RW performed on individual Gaussians) and cluster RW using Euclidean distance under identical brush inputs. The reported metrics (IoU, accuracy, F1 score, and precision) are measured for the target region shown in Figure 4, and execution time is reported in seconds. Gaussian RW achieves strong segmentation accuracy in some cases (e.g., vortex), but shows inconsistent performance across datasets and incurs significantly higher computational cost, requiring up to two orders of magnitude longer execution time than our method. In contrast, cluster RW using Euclidean distance provides faster execution, but leads to noticeably lower accuracy, particularly in IoU and F1 score. This highlights the importance of structure-aware distance, as Euclidean distance fails to capture underlying spatial connectivity in volumetric data. Our method consistently achieves high segmentation quality across all datasets while maintaining efficient runtime, demonstrating a better balance between accuracy and efficiency.

F TEXT LABEL LATENCY

We report the latency of real-time text labeling in Table 3. While user-perceived latency may vary with system conditions, we observe a consistent trend: latency increases slightly as the number of selected points increases. This is expected, as the selected Gaussians are rasterized into 2D before extracting CLIP image embeddings, and the cost of this 2D rasterization step scales with the number of selected Gaussians.

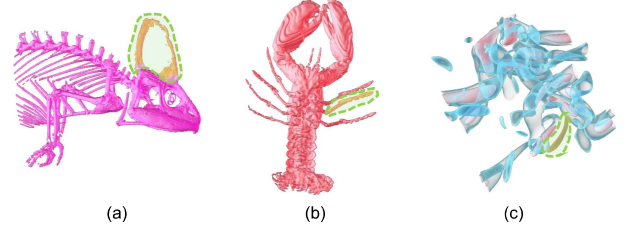


Fig. 4: Datasets used in the RW ablation: (a) chameleon, (b) lobster, and (c) vortex. The target selected is highlighted in a green dotted line.

Table 2: Comparison of RW variants across datasets.

Dataset	Method	Time (s)	IoU	Acc.	F1	Prec.
Chameleon	Ours	0.0807	0.8814	0.9977	0.8814	1.0000
	Cluster RW (Euclidean)	0.0217	0.5538	0.9854	0.5538	1.0000
	Gaussian RW (Geodesic)	9.9769	0.5327	0.9850	0.5327	1.0000
Lobster	Ours	0.0833	0.9012	0.9980	0.9018	0.9993
	Cluster RW (Euclidean)	0.0205	0.5538	0.9854	0.5538	1.0000
	Gaussian RW (Geodesic)	11.5861	0.8622	0.9971	0.8622	1.0000
Vortex	Ours	0.6044	0.5247	0.9850	0.6883	1.0000
	Cluster RW (Euclidean)	0.0903	0.1036	0.7273	0.1877	0.1036
	Gaussian RW (Geodesic)	4.7902	1.0000	1.0000	1.0000	1.0000

Consequently, the rendering overhead introduced during this preprocessing stage appears to be the primary factor affecting latency. The 2D rasterization and CLIP embedding extraction are performed on a local Python server. All components run on a single local machine, and communication between the PC-VR client and the Python server is handled via a RESTful API over the local loopback interface.

Table 3: Text labeling latency (in seconds) with respect to the number of selected Gaussians.

dataset	10 ²	10 ³	10 ⁴
chameleon	12.5	13.8	15.0
beetle	11.9	13.7	14.5
lobster	11.4	12.8	13.8
Nyx	11.2	12.1	13.9
aneurysm	11.2	11.8	12.7

G PER-PARTICIPANT RESULTS

From Tables 4 to 6, we report per-participant user study results for the three datasets where the completion time is in seconds, and TP, FP, FN, and TN are reported as the number of Gaussians.

H POST-INTERVIEW AND DISCUSSION

The post-interview responses provide additional insights into how participants perceived the selection workflow and interaction design. A consistent pattern was the use of a coarse-to-fine strategy, in which broad regions were first selected using RW or cluster mode and then refined using point mode selection (P2, P4, P7, P9, P10, P11). This behavior aligns with H2, suggesting that the hierarchical select-and-refine workflow effectively reduces the need for exhaustive point-based interaction while maintaining flexibility for fine-grained editing. This pattern was also explicitly described by participants. For instance, P9 noted that they often used “RW to first select a large region and then ‘cluster’ for refinement,” while P8 described a structured workflow where “RW → cluster → point” was used to progressively refine the selection and balance efficiency and accuracy.

Participants also emphasized that the choice of selection method strongly depended on dataset characteristics (P1, P2, P4, P7, P8, P9). Broad, clearly separated regions were typically handled using RW mode, whereas dense or structurally complex regions required cluster or point-based refinement. As P4 described, simple regions could be handled with point-based interaction alone. In contrast, more complex structures required combinations such as (cluster+point) or (RW+point) for efficient selection. Similarly, P8 reported that visually distinct

Table 4: Per-participant user study results for the vortex dataset.

method	pcp.	ACC	MCC	F1	IoU	prec.	int. count	comp. time	TP	FP	FN	TN
SuperSplat	P1	0.9883	0.8635	0.8655	0.7628	0.7923	16	40.63	4,294	1,126	209	108,815
GSVR	P1	0.9859	0.7953	0.7890	0.6515	0.9584	15	67.64	3,019	131	1,484	109,810
Super-Gaussian	P1	0.9903	0.8651	0.8671	0.7654	0.9429	9	238.30	3,614	219	889	109,722
SuperSplat	P2	0.9846	0.7735	0.7579	0.6102	0.9910	10	166.64	2,763	25	1,740	109,916
GSVR	P2	0.9828	0.7441	0.7282	0.5725	0.9655	19	283.57	2,632	94	1,871	109,847
Super-Gaussian	P2	0.9935	0.9106	0.9120	0.8382	0.9745	7	38.45	3,859	101	644	109,840
SuperSplat	P3	0.9910	0.8743	0.8737	0.7757	0.9767	18	138.21	3,559	85	944	109,856
GSVR	P3	0.9900	0.8588	0.8556	0.7476	0.9878	11	98.22	3,398	42	1,105	109,899
Super-Gaussian	P3	0.9967	0.9559	0.9574	0.9184	0.9736	2	17.44	4,241	115	262	109,826
SuperSplat	P4	0.9860	0.7969	0.7895	0.6523	0.9665	11	123.40	3,005	104	1,498	109,837
GSVR	P4	0.9893	0.8487	0.8478	0.7358	0.9612	3	11.26	3,415	138	1,088	109,803
Super-Gaussian	P4	0.9907	0.8701	0.8696	0.7693	0.9723	5	5.75	3,542	101	961	109,840
SuperSplat	P5	0.9954	0.9370	0.9388	0.8846	0.9740	31	113.08	4,080	109	423	109,832
GSVR	P5	0.9918	0.8852	0.8845	0.7929	0.9850	6	57.27	3,614	55	889	109,886
Super-Gaussian	P5	0.9991	0.9885	0.9889	0.9781	0.9781	6	24.47	4,503	101	0	109,840
SuperSplat	P6	0.9982	0.9768	0.9777	0.9563	0.9776	12	124.88	4,403	101	100	109,840
GSVR	P6	0.9746	0.7141	0.7203	0.5629	0.6363	5	48.26	3,737	2,136	766	107,805
Super-Gaussian	P6	0.9976	0.9674	0.9686	0.9392	0.9772	4	34.47	4,324	101	179	109,840
SuperSplat	P7	0.9982	0.9766	0.9774	0.9557	0.9625	14	105.32	4,470	174	33	109,767
GSVR	P7	0.9959	0.9446	0.9462	0.8979	0.9762	8	56.22	4,134	101	369	109,840
Super-Gaussian	P7	0.9972	0.9620	0.9634	0.9294	0.9769	3	9.32	4,279	101	224	109,840
SuperSplat	P8	0.9931	0.9044	0.9055	0.8273	0.9759	37	72.99	3,803	94	700	109,847
GSVR	P8	0.9895	0.8519	0.8488	0.7373	0.9822	17	103.79	3,365	61	1,138	109,880
Super-Gaussian	P8	0.9991	0.9885	0.9889	0.9781	0.9781	11	19.15	4,503	101	0	109,840
SuperSplat	P9	0.9988	0.9847	0.9852	0.9709	0.9709	17	132.78	4,503	135	0	109,806
GSVR	P9	0.9909	0.8722	0.8718	0.7728	0.9724	6	109.63	3,558	101	945	109,840
Super-Gaussian	P9	0.9970	0.9595	0.9610	0.9248	0.9768	6	19.24	4,258	101	245	109,840
SuperSplat	P10	0.9871	0.8159	0.8156	0.6886	0.9323	63	130.70	3,264	237	1,239	109,704
GSVR	P10	0.9904	0.8650	0.8642	0.7609	0.9720	8	56.06	3,503	101	1,000	109,840
Super-Gaussian	P10	0.9991	0.9885	0.9889	0.9781	0.9781	15	65.40	4,503	101	0	109,840
SuperSplat	P11	0.9908	0.8913	0.8926	0.8061	0.8233	3	44.31	4,389	942	114	108,999
GSVR	P11	0.9805	0.7027	0.6779	0.5128	0.9663	6	76.36	2,351	82	2,152	109,859
Super-Gaussian	P11	0.9884	0.8350	0.8318	0.7120	0.9701	2	4.19	3,278	101	1,225	109,840

regions were efficiently handled with RW selection alone, whereas regions with similar features required cluster-based selection followed by point-level refinement. These observations indicate that the framework supports adaptive interaction strategies that respond to structural complexity, separation, and feature similarity. Compared with traditional point-based selection approaches, participants consistently reported improved efficiency and reduced effort (P1, P4, P7, P8, P9, P10), supporting **H1**. Several participants noted that GSVR was time-consuming and physically demanding, while Super-Gaussian enabled faster selection through a combination of methods. In particular, P8 emphasized that “the difference in task speed and fatigue was noticeable,” and P10 noted that cluster-based interaction allows users to “select reasonably adjacent regions” efficiently. These responses highlight that the method’s benefit lies not only in accuracy but also in reducing repetitive manual interaction.

Participants also reported clear advantages of VR-based interaction over 2D approaches, particularly in terms of spatial understanding and interaction efficiency (P4, P5, P7, P8, P9, P10), supporting **H4**. VR allowed users to directly access and manipulate volumetric structures in 3D, reducing ambiguity caused by overlap and improving depth perception. For example, P5 noted that “In 2D-based interaction, I had to actively move and adjust the volume to understand its structure, while in VR, simply changing the viewing angle was enough to grasp it. Moreover, controlling the volume using only a mouse required a high level of concentration.” Similarly, P9 observed that “in the vortex dataset, it was much easier to find the desired region in VR than in 2D,” and that viewpoint control was significantly more flexible. P8 further emphasized that rotation, translation, and scaling in VR made it easier to understand spatial structures and perform analysis. However, we also observed that users with strong prior experience in 2D image editing tools (e.g., Photoshop) tended to remain comfortable with 2D-based workflows in certain scenarios (P3, P6), suggesting that familiarity can influence interaction preference.

The integration of NLI and text labeling was generally perceived as helpful for semantic interpretation and interaction (P3, P5, P6, P8, P9), supporting **H3**. Participants noted that NLI enabled intuitive querying and understanding of unfamiliar datasets. For instance, P3 highlighted that “I found NLI useful for obtaining explanations, as the system could provide explanations for specific regions or for the volume as a whole.”

Table 5: Per-participant user study results for the lobster dataset.

method	pcp.	ACC	MCC	F1	IoU	prec.	int. count	comp. time	TP	FP	FN	TN
SuperSplat	P1	0.8782	0.6825	0.7077	0.5477	0.5505	12	112.00	24,796	20,245	234	122,815
GSVR	P1	0.9899	0.9597	0.9655	0.9334	0.9798	15	85.42	23,821	492	1,209	142,568
Super-Gaussian	P1	0.9895	0.9605	0.9660	0.9342	0.9358	12	69.82	24,983	1,714	47	141,346
SuperSplat	P2	0.9912	0.9653	0.9705	0.9427	0.9707	6	53.73	24,286	732	744	142,328
GSVR	P2	0.9650	0.8566	0.8679	0.7666	0.9891	26	89.47	19,352	213	5,678	142,847
Super-Gaussian	P2	0.9955	0.9825	0.9851	0.9707	0.9792	6	21.76	24,808	528	222	142,532
SuperSplat	P3	0.9287	0.7623	0.7937	0.6580	0.6971	51	265.20	23,064	10,024	1,966	133,036
GSVR	P3	0.9804	0.9211	0.9318	0.8724	0.9645	43	184.00	22,559	830	2,471	142,230
Super-Gaussian	P3	0.9896	0.9594	0.9655	0.9333	0.9583	28	134.40	24,348	1,059	682	142,001
SuperSplat	P4	0.9828	0.9340	0.9437	0.8934	0.9187	17	109.67	24,280	2,148	750	140,912
GSVR	P4	0.9765	0.9052	0.9156	0.8443	0.9869	32	74.68	21,372	284	3,658	142,776
Super-Gaussian	P4	0.9939	0.9761	0.9796	0.9601	0.9748	20	63.74	24,643	638	387	142,422
SuperSplat	P5	0.9696	0.8770	0.8934	0.8074	0.9359	52	150.16	21,392	1,465	3,638	141,595
GSVR	P5	0.9702	0.8788	0.8903	0.8022	0.9861	14	34.78	20,309	286	4,721	142,774
Super-Gaussian	P5	0.9919	0.9676	0.9721	0.9457	0.9934	17	52.41	23,820	159	1,210	142,901
SuperSplat	P6	0.8754	0.6764	0.7024	0.5413	0.5450	13	51.53	24,718	20,634	312	122,426
GSVR	P6	0.9507	0.8413	0.8580	0.7513	0.7513	6	39.22	25,030	8,286	0	134,774
Super-Gaussian	P6	0.9251	0.7702	0.7947	0.6594	0.6714	5	42.79	24,370	11,929	660	131,131
SuperSplat	P7	0.9921	0.9687	0.9732	0.9478	0.9870	11	66.93	24,023	316	1,007	142,744
GSVR	P7	0.9869	0.9475	0.9543	0.9125	0.9904	29	92.03	23,045	224	1,985	142,836
Super-Gaussian	P7	0.9883	0.9532	0.9594	0.9220	0.9905	13	24.66	23,285	224	1,745	142,836
SuperSplat	P8	0.9775	0.9100	0.9231	0.8572	0.9380	113	348.04	22,742	1,502	2,288	141,558
GSVR	P8	0.9860	0.9441	0.9517	0.9079	0.9805	19	88.43	23,142	460	1,888	142,600
Super-Gaussian	P8	0.9923	0.9696	0.9740	0.9494	0.9843	51	162.81	24,127	384	903	142,676
SuperSplat	P9	0.9896	0.9589	0.9649	0.9322	0.9695	18	101.99	24,038	755	992	142,305
GSVR	P9	0.9912	0.9652	0.9701	0.9418	0.9890	32	168.07	23,824	265	1,206	142,795
Super-Gaussian	P9	0.9949	0.9803	0.9832	0.9669	0.9698	32	167.49	24,952	777	78	142,283
SuperSplat	P10	0.9705	0.8878	0.9044	0.8255	0.8728	76	126.31	23,489	3,424	1,541	139,636
GSVR	P10	0.9870	0.9483	0.9558	0.9153	0.9675	18	61.20	23,636	793	1,394	142,267
Super-Gaussian	P10	0.9882	0.9548	0.9613	0.9256	0.9387	23	98.16	24,658	1,611	372	141,449
SuperSplat	P11	0.9120	0.7217	0.7567	0.6086	0.6433	14	107.64	22,991	12,746	2,039	130,314
GSVR	P11	0.9792	0.9165	0.9282	0.8660	0.9538	8	38.74	22,626	1,097	2,404	141,963
Super-Gaussian	P11	0.9737	0.8998	0.9146	0.8427	0.8844	10	32.55	23,703	3,098	1,327	139,962

and preferred the scene editing through NLI. P6 reported that speech-based interaction was particularly useful when the text in the NLI panel was less readable. At the same time, several limitations were identified. Participants noted that latency in text labeling and LLM responses should be reduced.

Several usability-related limitations and improvement suggestions were also raised. Participants requested clearer visual feedback for brush interaction, such as indicating the brush tip or providing a preview (P5), and emphasized the need for undo functionality (P7). Others suggested improving controller-based interaction, including more intuitive button mappings and shortcut-based operations (P2, P8). For example, P2 suggested that brush activation should be toggleable via controller input with clear visual feedback, while P8 noted that mapping panel button inputs (e.g., mic button) to controllers could improve efficiency and bring VR interaction closer to the speed of traditional mouse-and-keyboard workflows.

Finally, participants highlighted that the proposed framework is particularly well-suited for complex, unfamiliar, or spatially overlapping datasets (P4, P5, P6, P7, P8, P10, P11). As P4 noted, “for unfamiliar and complex cases, it becomes easier to quickly understand and select the desired volume,” especially in datasets such as vascular structures or cosmological data. P8 further emphasized its applicability to medical datasets, noting that it can support fine-grained segmentation tasks such as distinguishing cancer and non-cancer regions within the same organ. Participants also suggested potential applications in other domains, including interior or structured scene datasets (P11).

Overall, the post-interview results reinforce the quantitative findings and support all four hypotheses.

Table 6: Per-participant user study results for the ionization dataset.

method	pcp.	ACC	MCC	F1	IoU	prec.	int. count	comp. time	TP	FP	FN	TN
SuperSplat	P1	0.9626	0.9250	0.9663	0.9348	0.9500	19	117.12	82,414	4,334	1,415	65,556
GSVR	P1	0.9256	0.8614	0.9268	0.8635	1.0000	11	34.54	72,389	2	11,440	69,888
Super-Gaussian	P1	0.9683	0.9383	0.9701	0.9418	1.0000	12	70.48	78,954	0	4,875	69,890
SuperSplat	P2	0.9760	0.9527	0.9775	0.9561	0.9977	12	95.14	80,328	189	3,501	69,701
GSVR	P2	0.4913	-0.0318	0.5468	0.3763	0.5318	58	185.38	47,180	41,543	36,649	28,347
Super-Gaussian	P2	0.9852	0.9706	0.9863	0.9729	0.9993	11	46.86	81,615	59	2,214	69,831
SuperSplat	P3	0.9796	0.9596	0.9810	0.9626	0.9982	4	57.84	80,832	142	2,997	69,748
GSVR	P3	0.9833	0.9665	0.9845	0.9696	0.9928	17	85.47	81,856	597	1,973	69,293
Super-Gaussian	P3	0.9804	0.9607	0.9820	0.9645	0.9883	10	44.25	81,788	965	2,041	68,925
SuperSplat	P4	0.9823	0.9650	0.9835	0.9675	0.9997	6	38.84	81,128	24	2,701	69,866
GSVR	P4	0.8934	0.8073	0.8917	0.8046	1.0000	6	22.99	67,448	2	16,381	69,888
Super-Gaussian	P4	0.9852	0.9706	0.9863	0.9729	0.9993	8	12.12	81,615	59	2,214	69,831
SuperSplat	P5	0.9626	0.9276	0.9645	0.9315	0.9991	22	65.74	78,151	68	5,678	69,822
GSVR	P5	0.9536	0.9094	0.9560	0.9157	0.9898	37	159.68	77,496	799	6,333	69,091
Super-Gaussian	P5	0.9867	0.9737	0.9877	0.9757	1.0000	56	180.49	81,792	0	2,037	69,890
SuperSplat	P6	0.9777	0.9560	0.9792	0.9592	0.9979	9	61.20	80,574	172	3,255	69,718
GSVR	P6	0.9565	0.9161	0.9586	0.9205	0.9978	4	23.91	77,316	169	6,513	69,721
Super-Gaussian	P6	0.9873	0.9745	0.9883	0.9768	0.9938	2	24.92	82,387	513	1,442	69,377
SuperSplat	P7	0.9843	0.9689	0.9854	0.9713	0.9990	4	30.15	81,500	80	2,329	69,810
GSVR	P7	0.9775	0.9555	0.9789	0.9587	0.9982	11	31.60	80,505	142	3,324	69,748
Super-Gaussian	P7	0.9818	0.9638	0.9830	0.9666	0.9983	2	9.50	81,167	142	2,662	69,748
SuperSplat	P8	0.9827	0.9656	0.9839	0.9684	0.9971	23	172.11	81,407	237	2,422	69,653
GSVR	P8	0.9772	0.9550	0.9786	0.9582	0.9990	14	70.50	80,401	83	3,428	69,807
Super-Gaussian	P8	0.9841	0.9686	0.9853	0.9709	0.9999	26	85.15	81,403	11	2,426	69,879
SuperSplat	P9	0.9873	0.9747	0.9882	0.9768	0.9989	6	44.36	81,966	88	1,863	69,802
GSVR	P9	0.8067	0.6730	0.7846	0.6455	1.0000	8	80.04	54,113	2	29,716	69,888
Super-Gaussian	P9	0.9912	0.9823	0.9918	0.9838	0.9993	5	6.66	82,529	59	1,300	69,831
SuperSplat	P10	0.9768	0.9542	0.9783	0.9576	0.9970	56	93.77	80,502	239	3,327	69,651
GSVR	P10	0.9610	0.9248	0.9630	0.9286	0.9998	8	13.05	77,855	16	5,974	69,874
Super-Gaussian	P10	0.9842	0.9688	0.9853	0.9711	1.0000	6	37.09	81,406	0	2,423	69,890
SuperSplat	P11	0.9668	0.9354	0.9686	0.9391	0.9998	5	96.62	78,732	12	5,097	69,878
GSVR	P11	0.9545	0.9128	0.9565	0.9166	0.9998	8	57.93	76,849	14	6,980	69,876
Super-Gaussian	P11	0.9807	0.9611	0.9822	0.9650	0.9871	9	24.93	81,929	1,071	1,900	68,819